

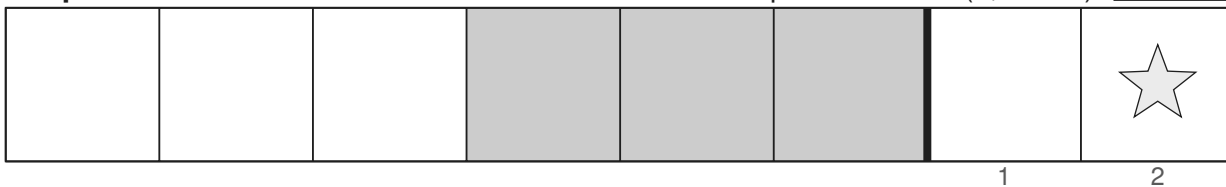
Problem 1: Put a penny on each gray square of strip A. The thick line is the river. A jump: a peg hops over a peg right next to it, in a straight line, into the empty square just beyond, and the peg it hopped over comes off the strip. Make one jump so that a peg crosses the river. Write an X in the square where it lands.

Strip A

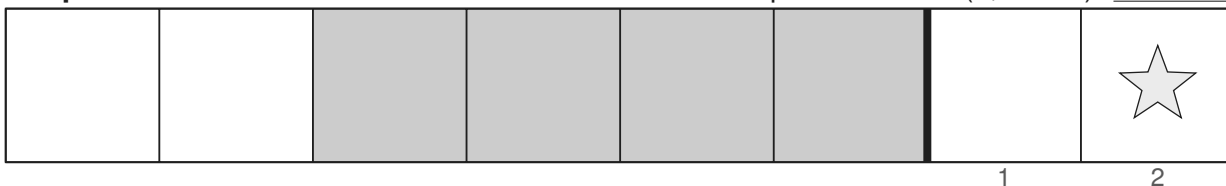
Problem 2: Strips B, C, D and E start with three, four, four and six pegs on the home side of the river. For each strip, put pennies on the gray squares, then jump as many times as you like, in either direction, and try to get a peg onto square 2, the square with the star. In the blank above the strip write the farthest square any peg reached: 0, 1 or 2, where 0 means that no peg crossed the river.

Strip B

farthest square reached (0, 1 or 2): _____

**Strip C**

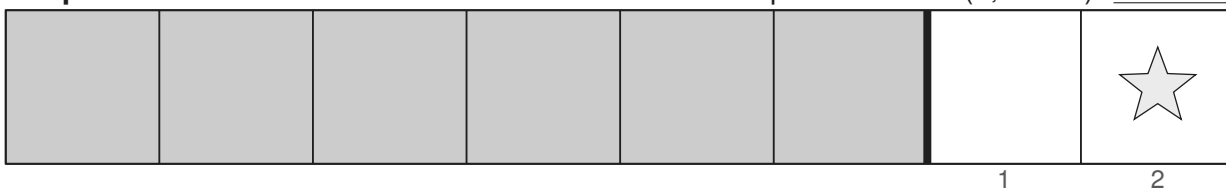
farthest square reached (0, 1 or 2): _____

**Strip D**

farthest square reached (0, 1 or 2): _____

**Strip E**

farthest square reached (0, 1 or 2): _____



Problem 3: On strips F1, F2 and F3, choose your own starting squares on the home side, up to six pennies, a different set on each strip. Draw a dot in each square you used, then play and write the farthest square reached, 0, 1 or 2.

Strip F1

farthest square reached (0, 1 or 2): _____

							★
						1	2

Strip F2

farthest square reached (0, 1 or 2): _____

							★
						1	2

Strip F3

farthest square reached (0, 1 or 2): _____

							★
						1	2

Problem 4: Strip G has a value printed in the corner of each square. The total of a position is the sum of the values of the squares that have pegs on them. For each row of the table: put pennies on the squares listed, write the total, make the jump described, and write the new total. In the last column write *up*, *same* or *down*.

Strip G

1	1	2	3	5	8	13	21 ★
						1	2

pegs on the squares valued . . .	total before	total after	up, same or down
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all six home squares; make the only jump that is possible	_____	_____	_____
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2 and 3; jump to the left	_____	_____	_____
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2 and 3; jump to the right	_____	_____	_____
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5 and 8; jump to the right, across the river	_____	_____	_____
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the second 1 and 2; jump to the left	_____	_____	_____
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8 and 13, one on each side of the river; jump to the left	_____	_____	_____
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Problem 5: Write the total of the six home squares of strip G: _____ and the value of square 2: _____. Strip H has seven home squares and no values. Write 1 and 1 in the corners of the first two home squares, then continue to the right: each value is the sum of the two values before it, up to the star. Write the total of the seven home squares: _____ and the value of square 2: _____. Can a peg ever land on square 2 of strip G? Tell the adult at your table how you know, and whether the same reason works on strip H.

Strip H

									★
								1	2

Problem 6: Board O is a grid with the river across it. Squares are named by column letter and row number, like C3. On a board, jumps go up, down, left or right, never diagonally. Put pennies on C3 and C2, make the one jump that crosses the river, and write the name of the square where the peg lands: _____

Problem 7: Use board O again. Put only three pennies on home squares of your choice and try to get a peg to the star at C5. Try two placements. Record each one on the small boards beside board O: draw a dot in each square you used and write the highest row number any peg reached.

Board O

			★		
5					
4					
3					
2					
1					
	A	B	C	D	E

Try 1

		★		
5				
4				
3				
2				
1				
	A	B	C	D

highest row: _____

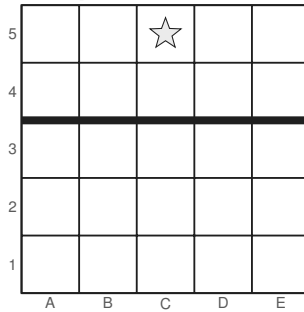
Try 2

		★		
5				
4				
3				
2				
1				
	A	B	C	D

highest row: _____

Problem 8: Find a shape of four pennies on home squares of board O (on page 3) from which a peg can reach the star at C5. Draw it on a small board with dots, write *yes* beside *works*, and list its jumps in order, like C1 over C2 to C3. Find a second shape that works. Then draw two shapes of four pennies that do not work, write *no* beside them, and for one of them tell the adult at your table why it cannot work.

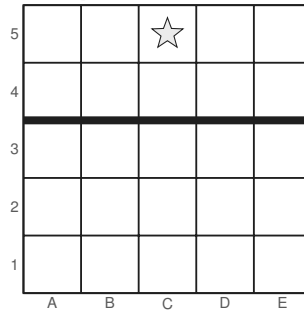
Shape 1



works: _____

jumps: _____

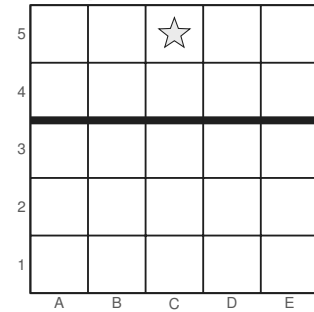
Shape 2



works: _____

jumps: _____

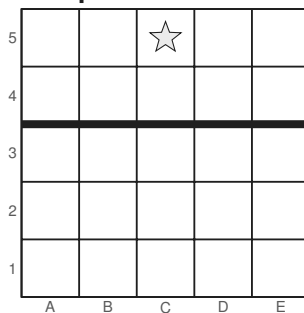
Shape 3



works: _____

jumps: _____

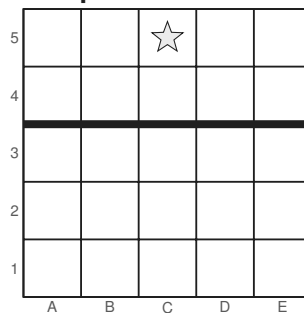
Shape 4



works: _____

jumps: _____

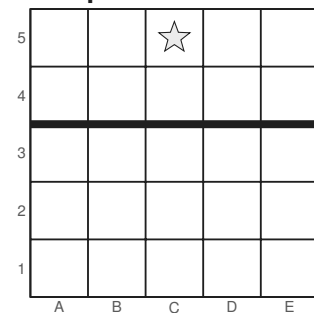
Shape 5



works: _____

jumps: _____

Shape 6



works: _____

jumps: _____

Problem 9: Board P shows a value in the corner of every square of board O. For each line, put pennies on the squares named, make the jump, and write the total before and the total after.

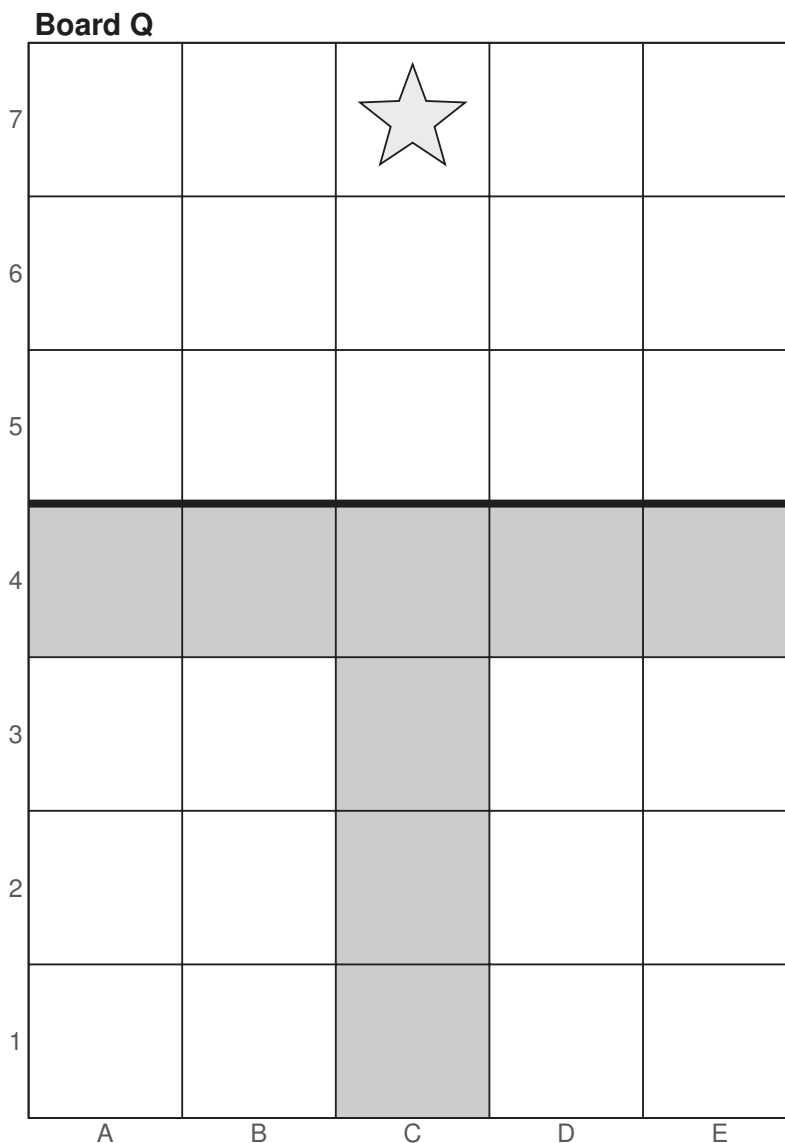
C3 and C2; jump up	before _____	after _____
D3 and E3; jump to the left	before _____	after _____
B3 and C3; jump to the right	before _____	after _____
C4 and C3; jump down	before _____	after _____

Board P

8	13	21	13	8
5				
5	8	13	8	5
4				
3	5	8	5	3
3				
2	3	5	3	2
2				
1	2	3	2	1
1				
A	B	C	D	E

Problem 10: On board P, find the three home squares with the largest values and write their total: _____. Write the value of the star: _____. Write the total of the four squares of a shape that worked in Problem 8: _____ and of the block C2, D2, C3, D3: _____. Put pennies on the block on board O and check whether a peg can reach the star. Tell the adult at your table what the totals show about Problem 7, and what the block shows about totals.

Problem 11: Board Q has eight pegs: five along the river and three below the middle one. Put pennies on the gray squares. Find seven jumps that put a peg on the star at C7. Write down each jump as you make it: the square the peg jumps from, the square it jumps over, and the square it lands on. If you get stuck, set the pegs up again and write a fresh list.



Jump 1: from _____ over _____ to _____

Jump 5: from _____ over _____ to _____

Jump 2: from _____ over _____ to _____

Jump 6: from _____ over _____ to _____

Jump 3: from _____ over _____ to _____

Jump 7: from _____ over _____ to _____

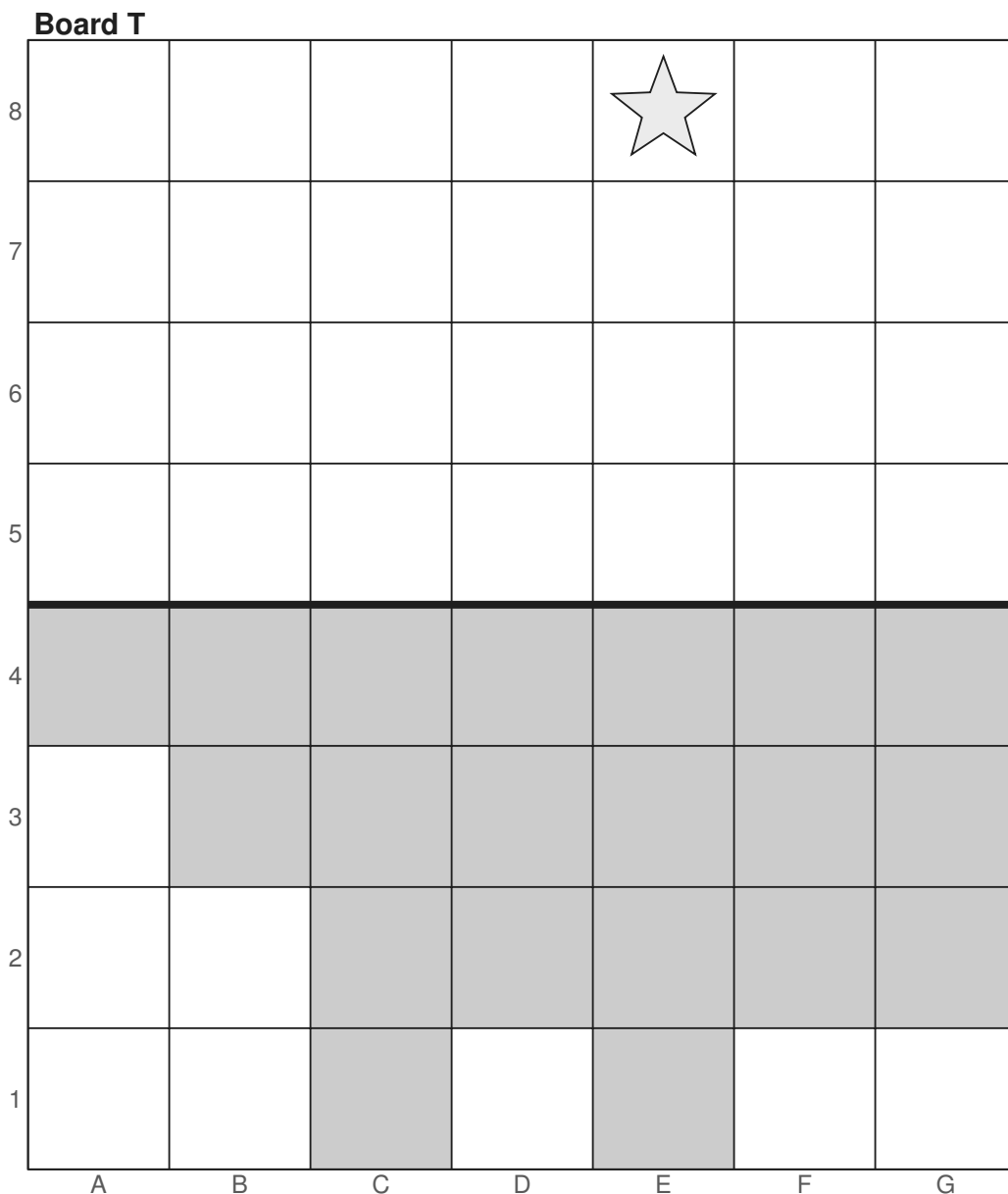
Jump 4: from _____ over _____ to _____

Problem 12: Board R shows the values that go with the star at C7. Write the total of the seven most valuable home squares: _____ and the value of the star: _____. Then write the total of the eight gray squares of board Q: _____. Say what these three numbers tell you about armies of seven pegs and about the army of board Q.

Board R

7	21	34	55 	34	21
6	13	21	34	21	13
5	8	13	21	13	8
4	5	8	13	8	5
3	3	5	8	5	3
2	2	3	5	3	2
1	1	2	3	2	1
	A	B	C	D	E

Problem 13: Board T on the next page has twenty pegs, and its star is in the fourth row across the river, at E8. Put pennies on the gray squares and find nineteen jumps that put a peg on the star. Before you start, decide which jump will be the last one, and which two squares must hold pegs just before it, and write that jump on the line above the jump list. Then write the jumps down as you make them. If you get stuck, set the pegs up again and write a fresh list.



Last jump: from _____ over _____ to _____

Jump 1: from _____ over _____ to _____

Jump 2: from _____ over _____ to _____

Jump 3: from _____ over _____ to _____

Jump 4: from _____ over _____ to _____

Jump 5: from _____ over _____ to _____

Jump 6: from _____ over _____ to _____

Jump 7: from _____ over _____ to _____

Jump 8: from _____ over _____ to _____

Jump 9: from _____ over _____ to _____

Jump 10: from _____ over _____ to _____

Jump 11: from _____ over _____ to _____

Jump 12: from _____ over _____ to _____

Jump 13: from _____ over _____ to _____

Jump 14: from _____ over _____ to _____

Jump 15: from _____ over _____ to _____

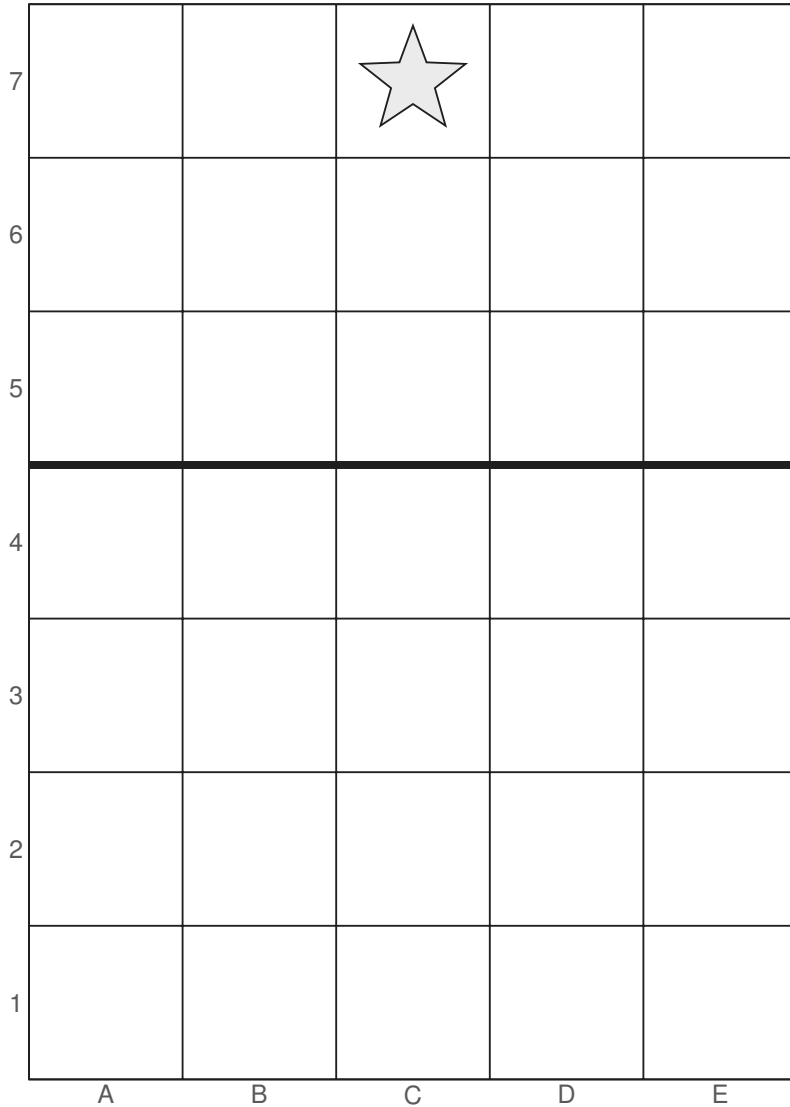
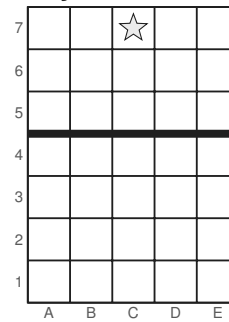
Jump 16: from _____ over _____ to _____

Jump 17: from _____ over _____ to _____

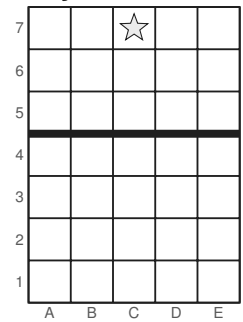
Jump 18: from _____ over _____ to _____

Jump 19: from _____ over _____ to _____

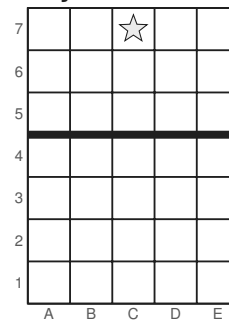
Problem 14: Board S is empty. Find a set of eight home squares, different from the shape of board Q, from which a peg can reach the star at C7. Test each set with pennies on board S. Record each set you try on one of the small boards: a dot in each square, and the highest row reached. When a set works, draw a dot in each of its eight squares on board S and list its seven jumps.

Board S**Try 1**

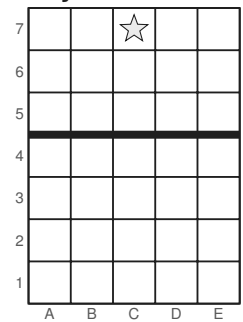
highest row: ____

Try 2

highest row: ____

Try 3

highest row: ____

Try 4

highest row: ____

Jump 1: from ____ over ____ to ____

Jump 5: from ____ over ____ to ____

Jump 2: from ____ over ____ to ____

Jump 6: from ____ over ____ to ____

Jump 3: from ____ over ____ to ____

Jump 7: from ____ over ____ to ____

Jump 4: from ____ over ____ to ____